

A high-angle, black and white photograph of a massive concrete dam. The dam's surface is composed of large, rectangular concrete panels. A narrow walkway with a metal railing runs along the top edge of the dam. A small figure of a person stands on this walkway, providing a sense of scale to the enormous structure. The sky is a uniform, dark grey.

MASTER SEMINAR:
SELECTED TOPICS IN MACHINE LEARNING RESEARCH

SEMINAR 1-SOICT 2026

SEMINAR HIGHLIGHTS

- ▶ **Discover** state-of-the-art ML research.
- ▶ Critically **analyse** recent publications.
- ▶ Advance your **scientific writing**.
- ▶ Practice conference-level **peer reviewing**.
- ▶ Enhance your **presentation skills**.



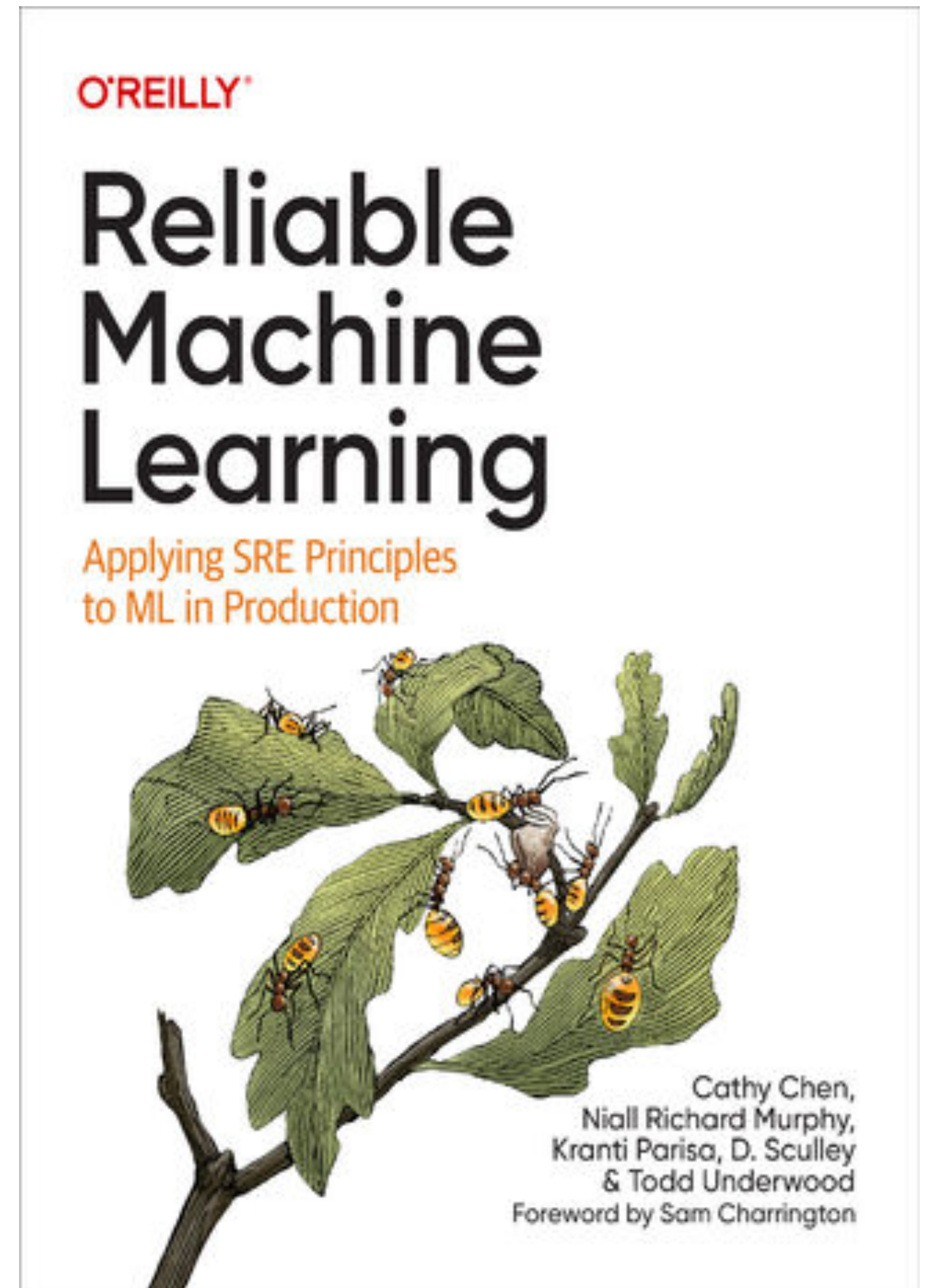
GRAPH NEURAL NETWORKS

- ▶ **GNN1: Spatial and Spectral Graph Neural Networks**
- ▶ **GNN2: Positional Encodings: From Sequences to Graphs**
- ▶ **GNN3: Unsupervised Learning on Graphs**
- ▶ **GNN4: Approaches to Link Prediction**
- ▶ **GNN5: Gaussian Processes on Graphs**
- ▶ **GNN6: Node Regression on Graphs**
- ▶ **GNN7: Kolmogorov-Arnold Networks & applications to graphs**
- ▶ **GNN8: Spectral Graph Neural Networks**
- ▶ **GNN9: Randomized Smoothing for Graphs**

RELIABLE MACHINE LEARNING

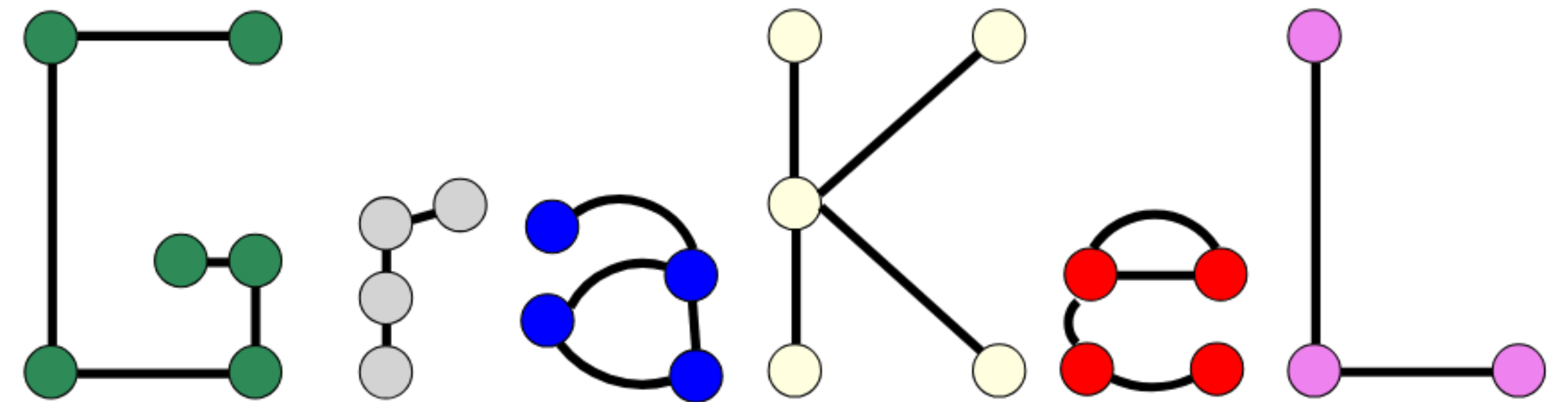
BUILDING AI AND PREDICTIVE SYSTEMS THAT ARE SAFE, ROBUST, AND DEPENDABLE IN REAL-WORLD ENVIRONMENTS.

- ▶ RML1: Robustness & Adversarial Defense
- ▶ RML2: Explainable AI (XAI)
- ▶ RML3: Uncertainty Quantification
- ▶ RML4: Fairness, Bias, & Alignment
- ▶ RML5: Differentially private time series analysis



KERNEL METHODS

- ▶ KM1: Kernel methods for time series
https://ceur-ws.org/Vol-2259/aics_7.pdf
- ▶ KM2: Kernel methods for graphs
<https://cs.stanford.edu/~quocle/srl-book.pdf>
- ▶



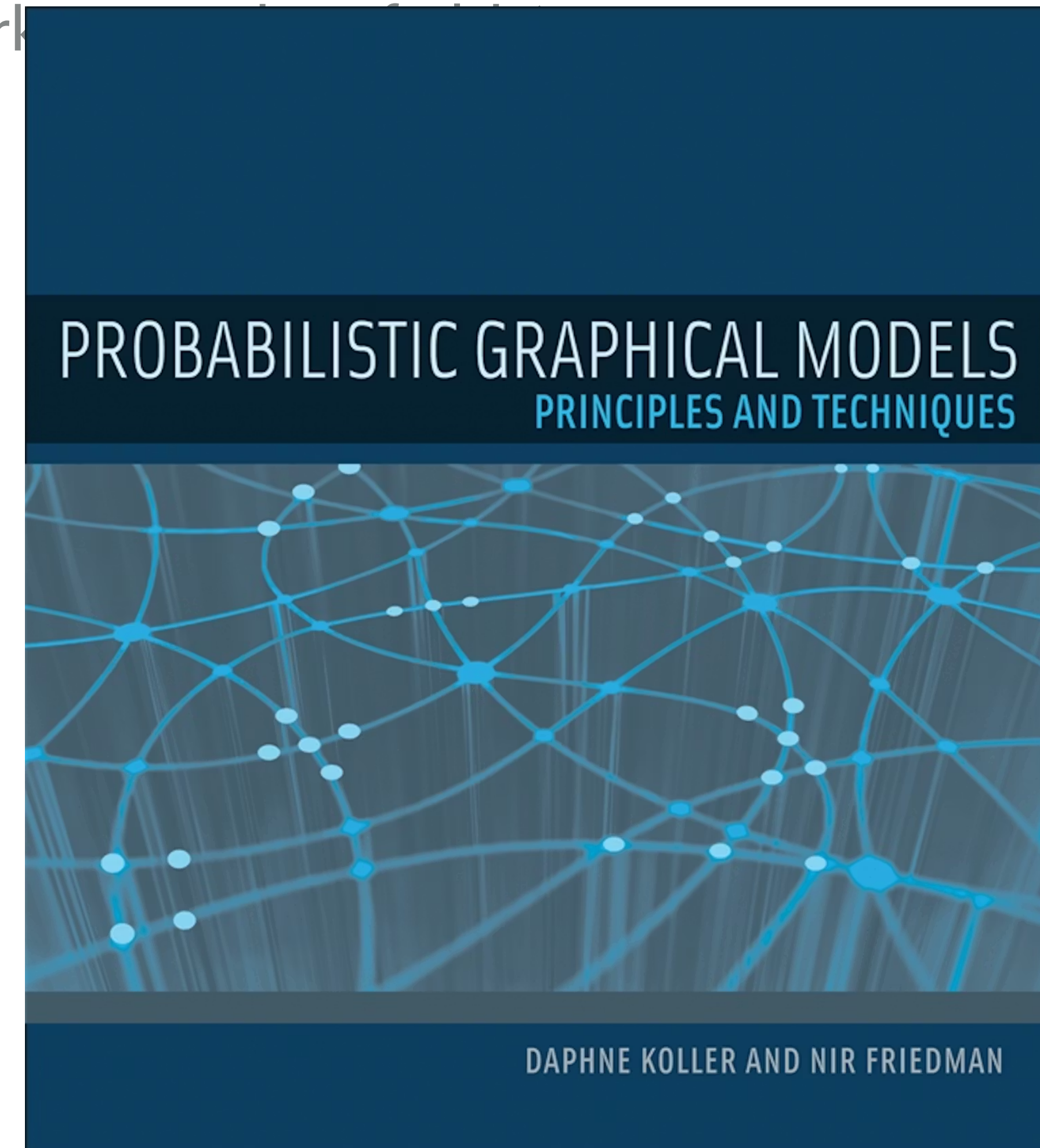
ROBUSTNESS, CERTIFICATION & EFFICIENCY

- ▶ RCE1: Certified Robustness against Poisoning Attacks
- ▶ RCE2: Inference Efficiency with Pruning/Quantization
- ▶ RCE3: Certified unlearning
- ▶ RCE4: Data Efficiency in Robust Training

GRAPHICAL PROBABILISTIC MODELS

<https://ermongroup.github.io/cs228-notes/>

- ▶ PGM1: Different representations (Bayesian Networks and Markov
- ▶ PGM2: Inference algorithms
- ▶ PGM3: Parameter learning
- ▶ PGM4: State sequences and hidden states
- ▶ PGM5: Sampling techniques
- ▶ PGM6: Dimensionality reduction
- ▶ PGM7: Model comparison
- ▶ PGM8: Continuous latent variables



OTHER TOPICS

- ▶ OT1: Bayesian Flow Networks
- ▶ OT2: Discrete Generative Models
- ▶ OT3: Schrödinger Bridge Matching
- ▶ OT4: Deep implicit models
- ▶ OT5: Unlearning for LLMs
- ▶ All other related topics proposed by the student and confirmed by the supervisor

WHAT WILL YOU DO?

- ▶ Select the topics and read the seed research papers
- ▶ Research:
 - ▶ Thoroughly understand and contextualise your topic (find relevant background references, potentially also related works)
 - ▶ Start your snowball research from there (references from / to these papers, relevant keywords)
- ▶ **Summarise** your findings, criticism and research ideas in a **short paper** (4-6 pages, double column)
- ▶ **Write reviews** for other students work
- ▶ **Present** your work in 40-minute talks
- ▶ Grades will be based on all parts: paper(20%), reviews(20%), talk(40%) & overall participation(20%).

SCHEDULE

- ▶ 02/06: Kick-off (Organisation and representation the list of topics)
- ▶ 09/06: Literature outline (presented by supervisor) and the topics proposed by students (presented by the authors)
- ▶ 16/06: Topic selection and confirmation, reviewer assignment
- ▶ 23, 30/6: Consultation meeting (Questions and Comments)
- ▶ 3/7: short paper submission
- ▶ 3/7, 6/7, 8/7, 25/7, 27/7, 30/7: Presentation
- ▶ 30/7: review submission